

Trainings ▾

General Information

All maintenance procedures listed for previous intervals **must** also be performed.

- Air Cleaner Element - Check
- Air Cleaner Precleaner - Check
- Air Cleaner Restriction - Check
- Air Intake Hoses, Pipes, and Clamps - Check
- Air Tanks - Drain
- Coolant Level - Check
- Drive Belts - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Belt Tension - Check
- Belt Tensioner - Check
- Fan Hub, Belt-Driven - Check
- Coolant Filter - Change
- Air Cleaner Element - Change
- Cooling Fan - Check
- Crankcase Breather Tube - Check
- Engine Wiring - Check
- Fuel Filters - Change (1)
- Lubricating Oil and Oil Filter - Change (2)
- Supplemental Coolant Additive - Check
- Batteries - Check
- Cold Start Aids (Seasonal) - Check
- Crankcase Breather (External) - Clean
- Crankshaft End Clearance - Check
- Engine Mounting Bolts - Check
- Hoses, Engine - Replace
- Steam Clean Engine - Clean
- Cooling System - Clean
- Fan Hub, Belt-Driven - Check
- Fan Idler Pulley Assembly - Check
- Vibration Damper - Check
- Water Pump - Check
- Fuel Injectors (all applications with Mechanically Actuated Fuel Injectors) - Replace

Notes:

1.

The replacement of the fuel filters can be extended to 500 hours, as long as the fuel inlet restriction does **not** exceed the maximum specified in Section V.

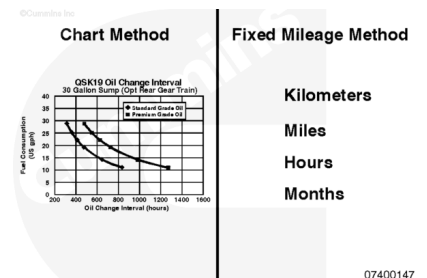
2.

The lubricating oil and filter change intervals can be extended, based on the engine's horsepower rating, oil pan capacity, and fuel consumption rate. Refer to the Oil Drain Intervals information in this section for specific details. Replacement of the oil filters **must not** be extended beyond 500 hours.

Oil Drain Intervals

There are two recommended methods to determine the proper oil and filter change interval:

- Chart Method (based on known fuel consumption rates).
- Fixed Mileage Method (based on fixed kilometers, miles, hours, or months; whichever occurs first)



Fixed Mileage Method:

**Table 2,
Fixed
Mileage
Method**

	52 liters [13.7 gal] pan			135 liters [35.6 gal] pan		
QSK23 Engine Rating	CF-4, CES 20075 Grade Oil	CG-4, CES 20075 Grade Oil	CH-4/CH-4+, CES 20071/CE S 20076 Grade Oil	CF-4, CES 20075 Grade Oil	CG-4, CES 20075 Grade Oil	CH-4/CH-4+, CES 20071/CE S 20076 Grade Oil
Up to 800 hp	250 hours	250 hours	500 hours	500 hours	500 hours	500 hours

Table 2, Fixed Mileage Method						
	52 liters [13.7 gal] pan			135 liters [35.6 gal] pan		
Above 800 hp	250 hours	250 hours	250 hours	500 hours	500 hours	500 hours
Note: Use the chart method and/or periodic oil sampling to determine if longer drain intervals are possible.						

Table 2, Fixed Mileage Method		
	52 liters [13.7 gal] pan	135 liters [35.6 gal] pan
Note: Standard oils will meet the requireme nts of CES 20075 (API CG-4 and CF-4). Premium oils will meet the requireme nts of CES 20071 (API CH-4) or CES 20076 (API CH- 4+). Refer to Section V for oil specificati ons.		

Chart Method:

Determine fuel and oil consumption rates:

- To use the Chart Method, accurate fuel consumption records **must** be kept and maintained.
- As fuel consumption rates change as a result of a change in operation or duty cycle of a particular engine, the oil change interval established by the Chart Method **must** be reevaluated, based on the change in fuel consumption.

Determine total lubricating oil system capacity:

- Total lubricating oil system capacity can be determined by adding the high level of the oil in the oil pan, plus the capacity of the two combination full-flow/bypass oil filters. See chart

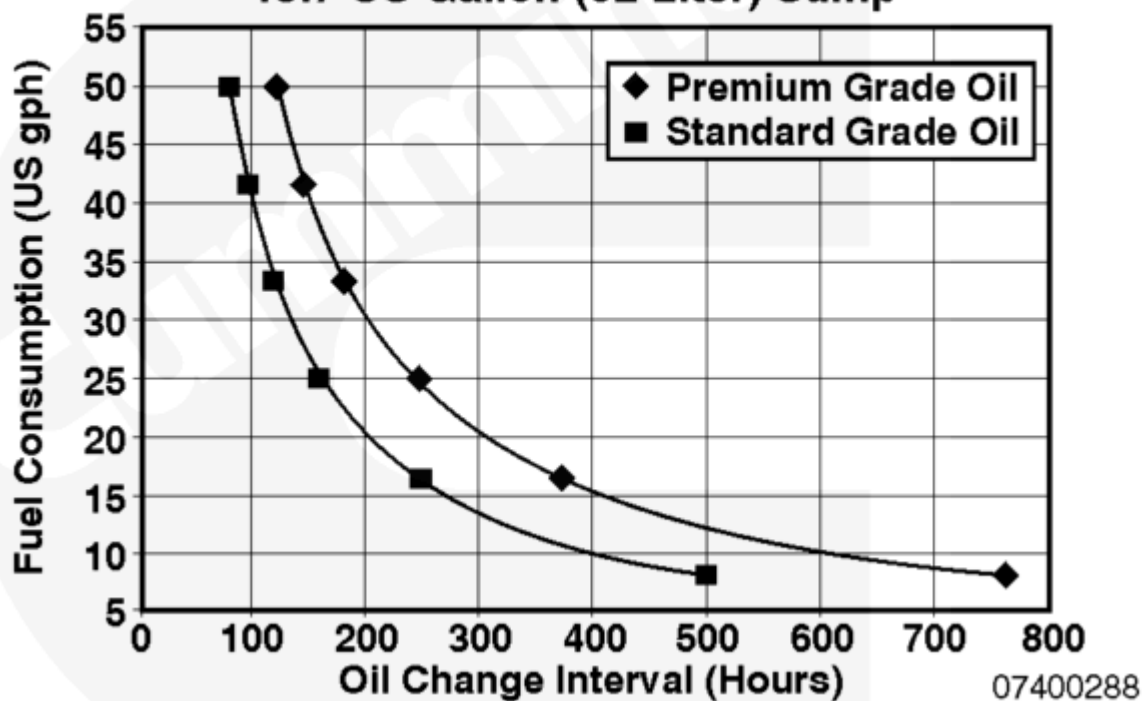
below.

**Table 3, QSK23
Sump Capacities**

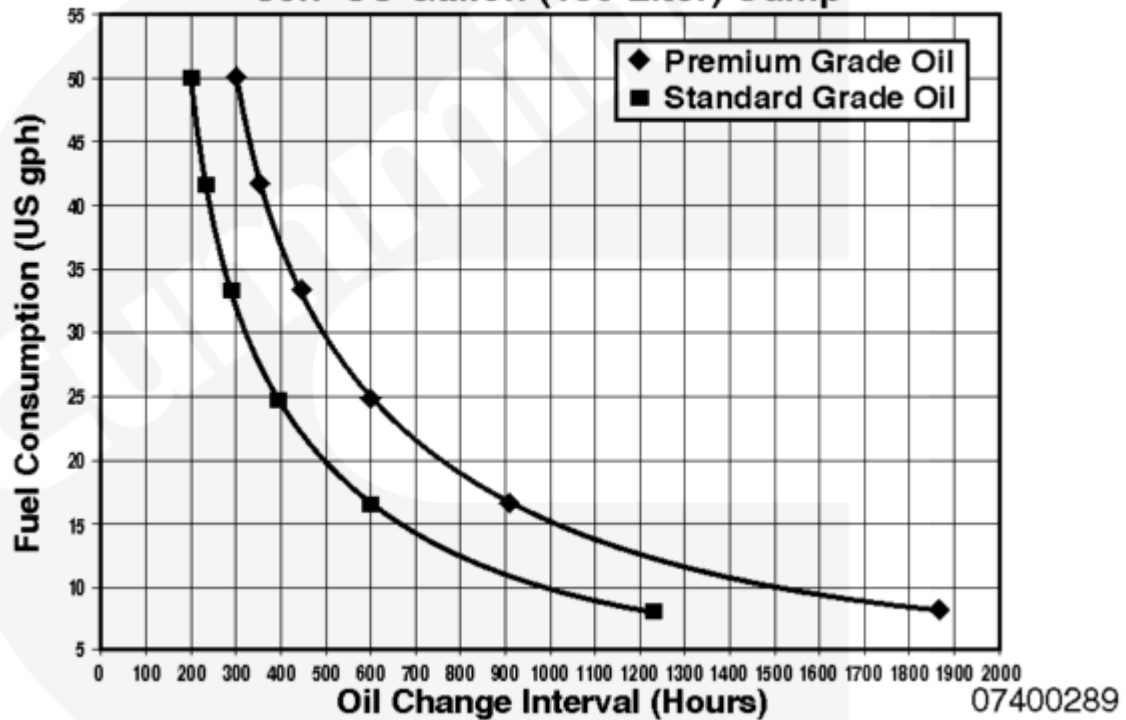
Rated Sump Capacity	Sump Capacity at Low Level Mark on Dipstick in Liters [Gal]	Sump Capacity at High Level Mark on Dipstick in Liters [Gal]	Total System Capacity (Sump and Two Combo Oil Filters) in Liters [Gal]
52 liters [13.7 gal]	42 liters [11.1 gal]	52 liters [13.7 gal]	57 liters [15.1 gal]
135 liters [35.7 gal]	66 liters [17.4 gal]	135 liters [35.7 gal]	140 liters [37 gal]

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**QSK23 Oil Change Interval
13.7 US Gallon (52 Liter) Sump**



QSK23 Oil Change Interval 35.7 US Gallon (135 Liter) Sump



To read the chart for premium grade oil:

Assume the engine has a 13.7-gallon sump capacity and the engine consumes 15 gallons of fuel per hour.

- Select the chart titled 13.7 gallons.
- The left vertical axis of the chart represents fuel consumption in gallons per hour.
- Determine the location of 15 gallons on the left vertical axis and draw a horizontal line across the chart until it intersects the curve with the diamonds.
- At the point of intersection, draw a straight vertical line down to the bottom of the graph.
- The point at which the vertical line intersects the bottom horizontal axis of the graph indicates the oil change interval (approximately 420 hours in this example).

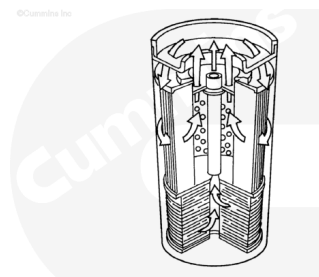
Premium oil filters should be used in conjunction with premium-grade oils when using the premium-grade oil curves to determine oil change intervals. Refer to Section V for lubricating oil filter specifications.

Premium filters contain synthetic media materials to provide more efficient filtration for the entire service life and extend the media life over conventional cellulose media.

Premium combination filters are made with synthetic media and have StrataPore™ designation on the outside of the filter.

StrataPore™ filters have the efficiency, capacity, and strength needed for extended service intervals.

The filter service intervals **must not** exceed 500 hours.



Last Modified: 04-Aug-2009
